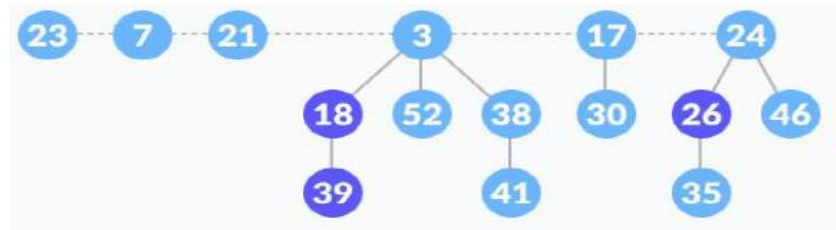
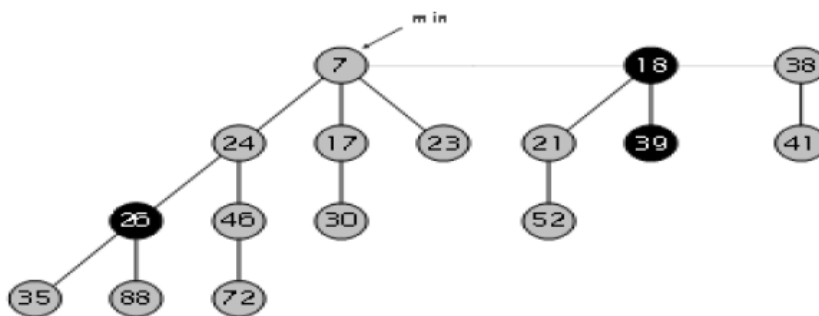


1. Design red black tree of the following sequence  
 90,130,143,151,123,133,164,157  
 Delete the following node from the above tree  
 151, 157, 143

2. Perform an extract min operation on the heap below



3. In the following Fibonacci heap decrease the key value of
  - i. 46 to 29 to 15
  - ii. 35 to 5



4. Let us consider a simple hash function as “key mod 7” and a sequence of keys as 51, 700, 76, 81, 94, 62, 111. Implement hashing and remove collisions using quadratic probing.
5. Design a perfect skip list for the given sequence 10, 40, 30, 50, 20, 15, 45, 35 and delete 30 and 45 from the resultant list.